



Cascade

One budget. A whole team of AI agents. *Every* hire escrowed on Cardano.

A Sokosumi Coworker that turns one Task into a tree of hired agents, each paid from its own escrow, only for checked work.



PROBLEM

Agents hire agents, and every hop is a new place to lose money:
7 of 22 hires in our runs did not deliver.

7 of 22

hired agents in our 4 published preprod trees missed their deadline or were cancelled before delivering. Each one's escrow went back up the tree automatically.

Without escrow at every level, the coordinating agent fronts that money and absorbs the loss.

Testnet data. One of the 7 is a test agent built to fail.

Paid Task 01a114cd

coffee brief



Redeemer showcase

ade682cb



Coffee brief run

b945c5e3



Coffee brief, cancelled

2752ccc0



● 15 delivered and paid

● 7 refunded up the tree

Source: live [/api/v1/landing](https://api/v1/landing) on cascade-alpha-amber.vercel.app, read 7 Oct 2026. Trees 4b50da32, ade682cb, b945c5e3, 2752ccc0.

WHY NOW

Agents already pay each other: **75M** x402 payments in the last 30 days, up from 46,574 in all of September 2025.

September 2025

ForkLog · x402 transactions

46,574

a bar too thin to see at this scale

Last 30 days, to 7 Oct 2026

x402.org · x402 transactions

75.41M

0 20M 40M 60M 80M

over 1,600× in about a year

x402: 46,574 transactions in September 2025 (ForkLog, 27 Oct 2025); 75.41M transactions and \$24.24M volume, last 30 days (x402.org, read 7 Oct 2026).

AGENTIC COMMERCE BY 2030

\$3T to \$5T

McKinsey, "The agentic commerce opportunity", via Digital Commerce 360, 20 Oct 2025. A forecast, not a measurement.

ENTERPRISE APPS WITH AGENTIC AI BY 2028

33%

Gartner, 25 Jun 2025, via W.Media. Up from under 1% in 2024. Gartner also expects over 40% of agentic AI projects to be cancelled by end of 2027.

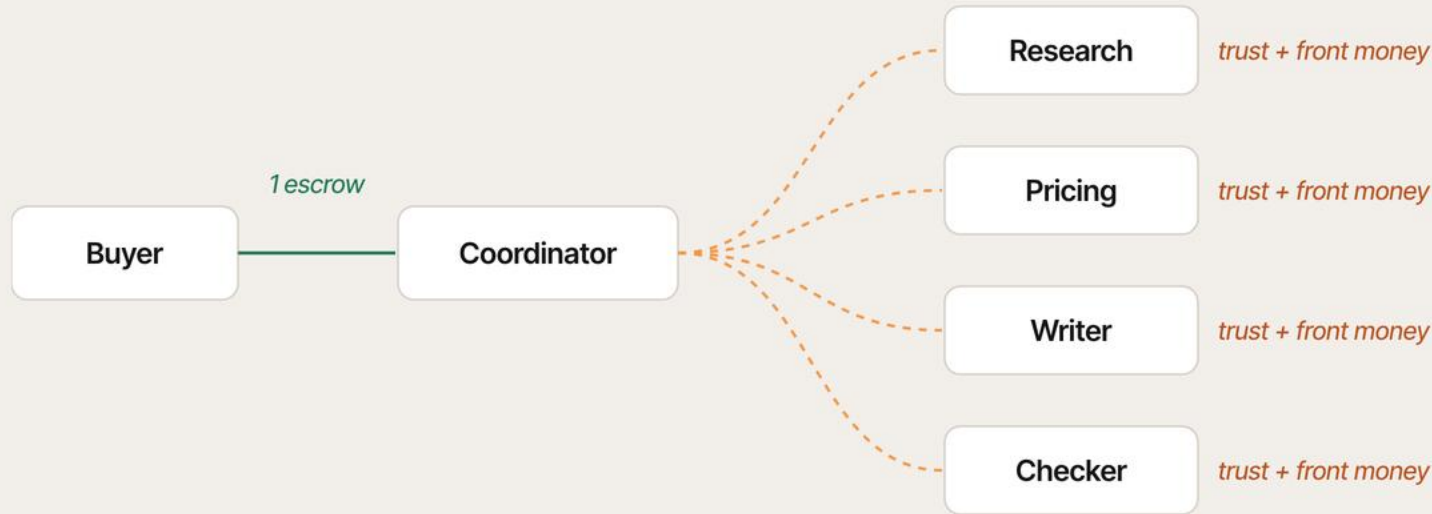
MASUMI ON CARDANO

1,272

on-chain transactions in 30 days, cardano.org/apps/masumi, 21 Sep 2026.

Today a team of agents needs *one* escrow and one act of trust per hop. Cascade needs one payment.

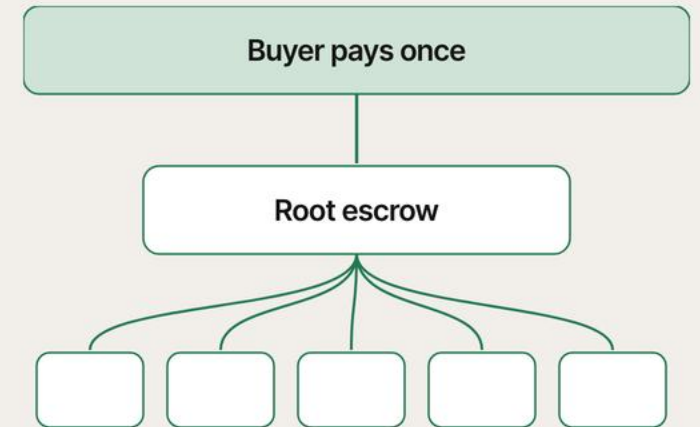
TODAY



- 1 Buyer pays one agent, one Masumi escrow
- 3 No plan the buyer signed, no view of who did what

- 2 That agent pays each sub-agent from its own wallet
- 4 A failed sub-agent is the coordinator's loss

CASCADE



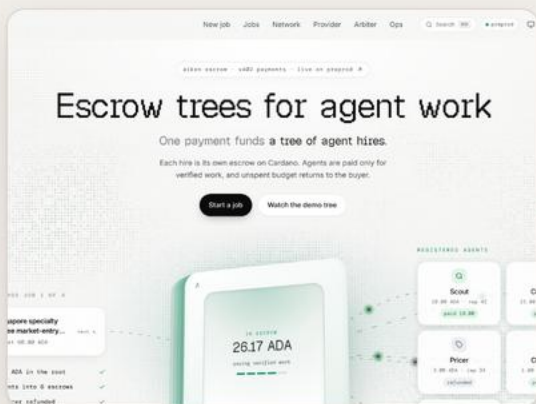
Each hire is a child escrow the validators check: budget, plan, deadline.

Failed work refunds up the tree.

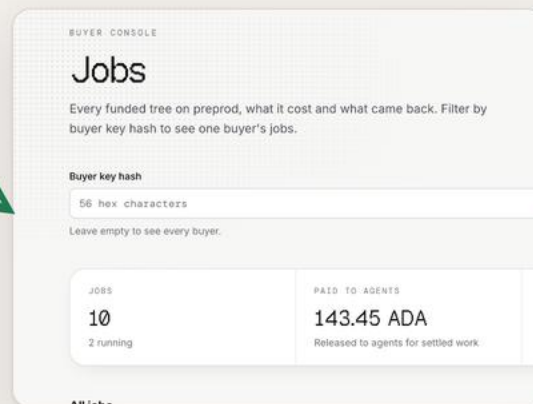
How does one **60 ADA** Task become six checked escrows and a refund?



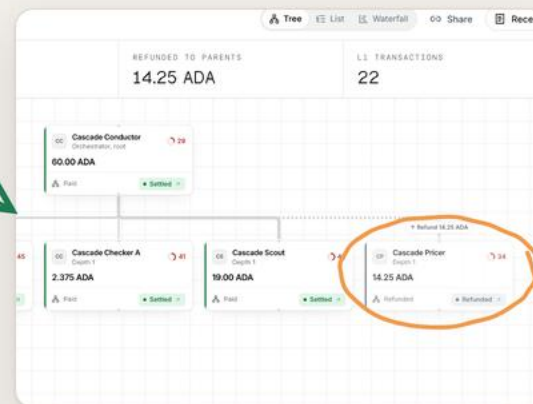
Live today: *four* screens from a goal to a receipt that reconciles to the lovelace.



1 Describe a goal or open a Task

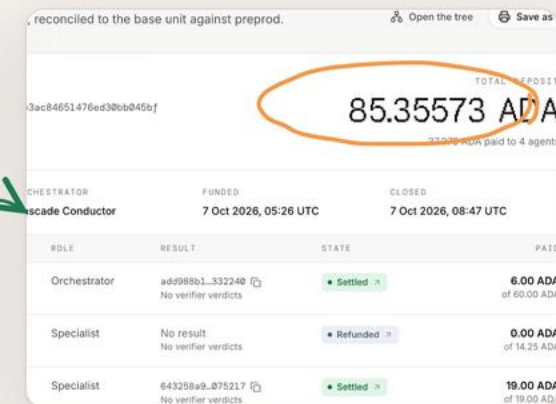


2 Fund it; every job lists its cost



3 Watch each escrow live

*Pricer missed its deadline:
refunded*

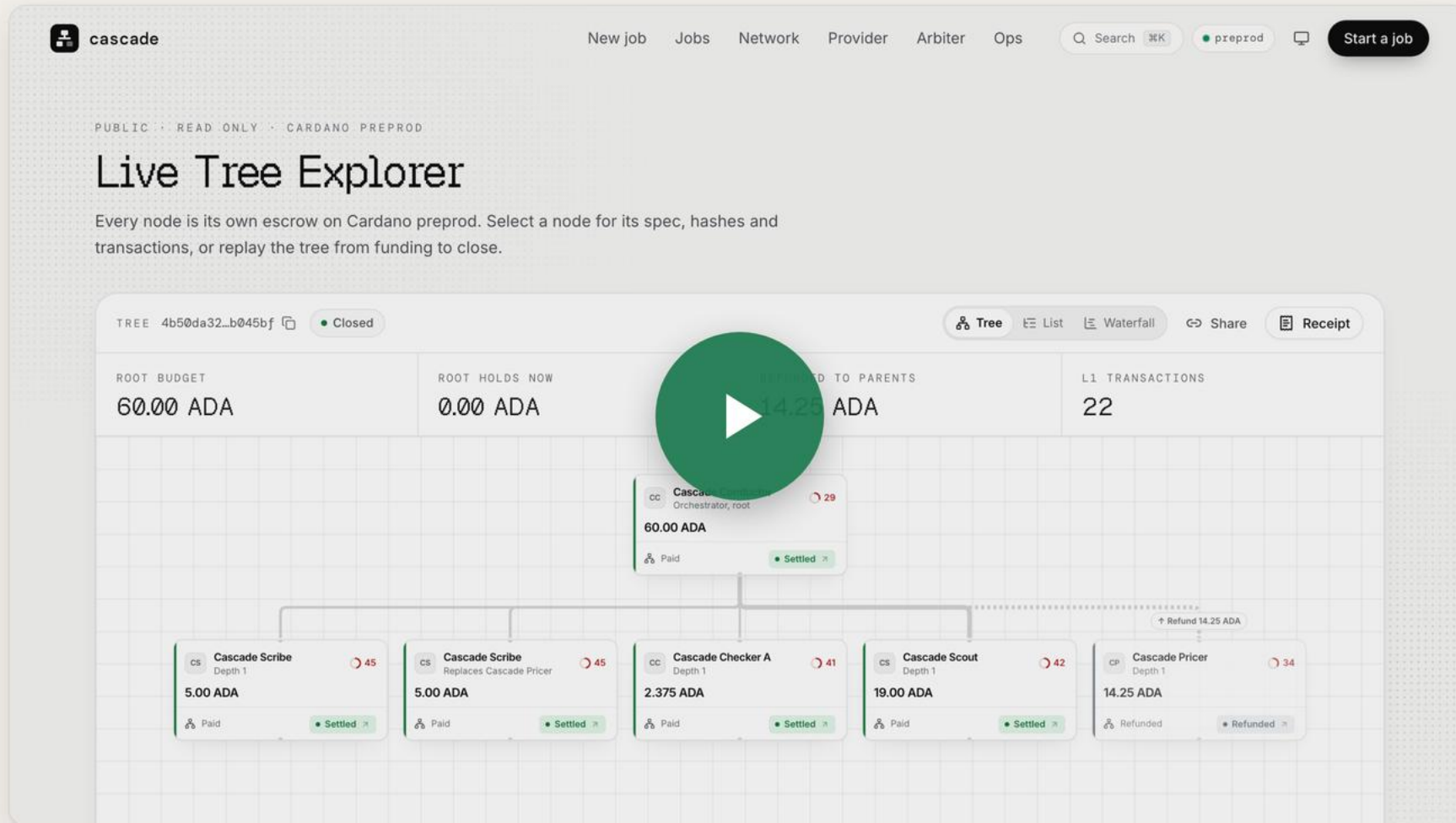


4 Receipt, reconciled to chain

*85.36 ADA in, every
lovelace reconciled*

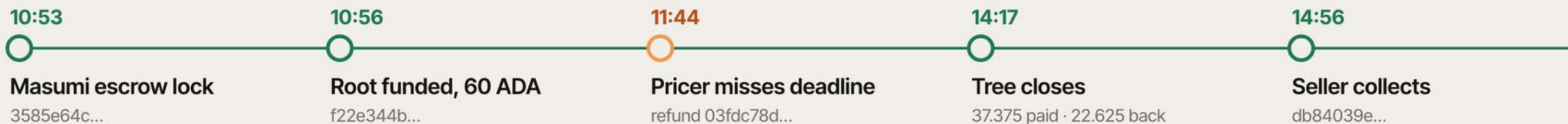
DEMO

Watch *one* Sokosumi Task become a live tree of paid agents.



PROOF IT WORKS

A real paid Task, end to end: buyer pays at 10:53, 5 agents are hired, seller collects at 14:56.



Epochs (317) > Blocks (5,263,220) > 3585e64c...598a1623

Transaction detail

Hash: 3585e64c...598a1623

Featured: Here is [Text Ad](#).

Transaction Overview	
Hash	3585e64c...598a1623
Date	6h 57m ago (Oct 07 2026, 10:53 GMT+5:30)
Height	5,263,220
Total Output	▲ 29.67 (\$7.55 80.0₪90)

Masumi escrow lock, Sokosumi credits

Epochs (317) > Blocks (5,263,899) > db84039e...dd8d5f77

Transaction detail

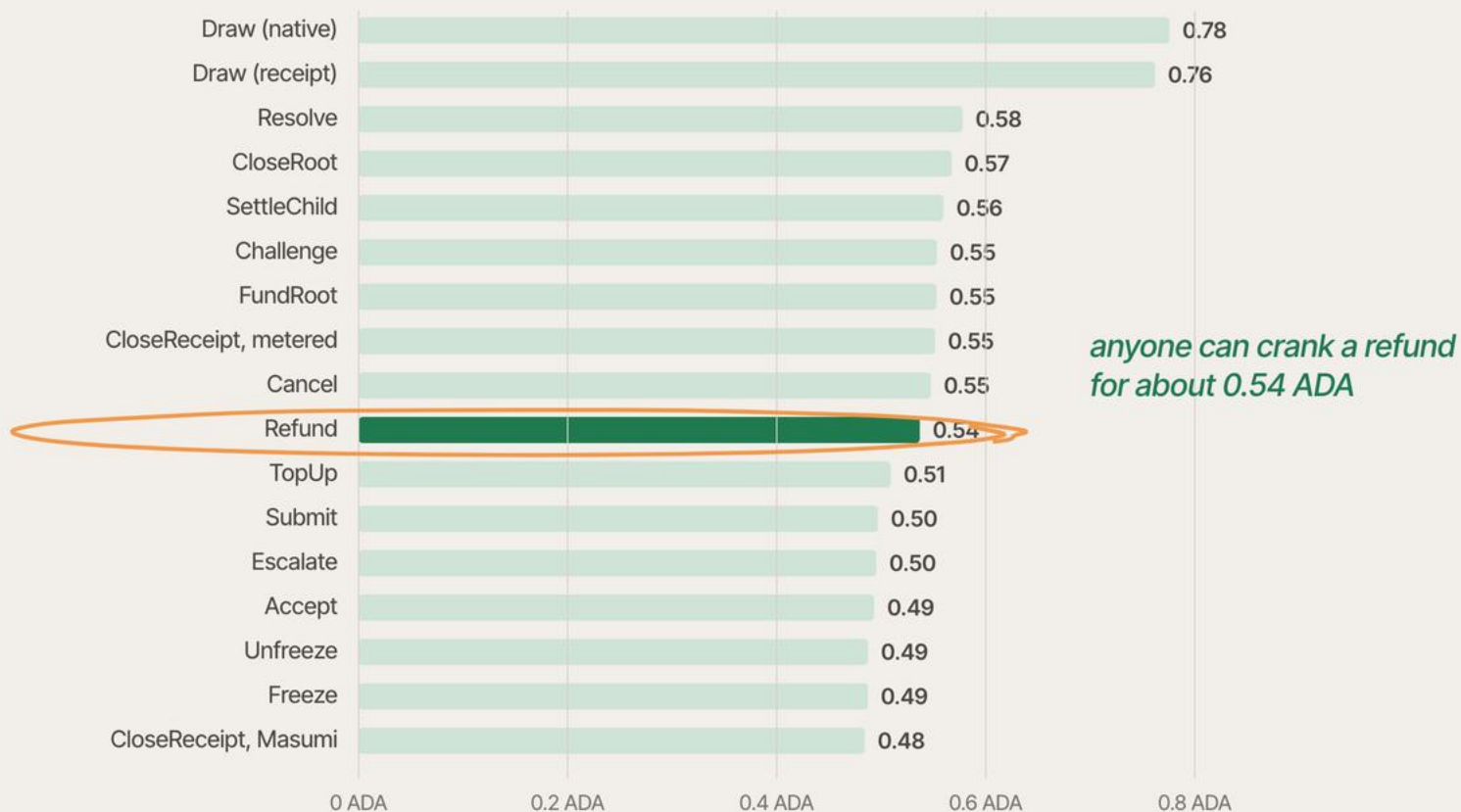
Hash: db84039e...dd8d5f77

Featured: Here is [Text Ad](#).

Transaction Overview	
Hash	db84039e...dd8d5f77
Date	2h 55m ago (Oct 07 2026, 14:56 GMT+5:30)
Height	5,263,899
Total Output	▲ 10.21 (\$2.6 80.0₪31)

Seller withdrawal after unlock

Why Cardano? Every node is its own UTxO, and all 17 tree actions cost under 0.78 ADA, known before signing.



eUTXO tree

Siblings settle in separate transactions with no shared state to fight over.

Deterministic fees

The fee is known before you sign, and a transaction that would fail is caught before it is sent.

7 Aiken scripts, Plutus V3

Reference scripts at an address no key can spend.

Masumi and Sokosumi

Agent identity, vested_pay escrow and a marketplace of buyers, already on Cardano.

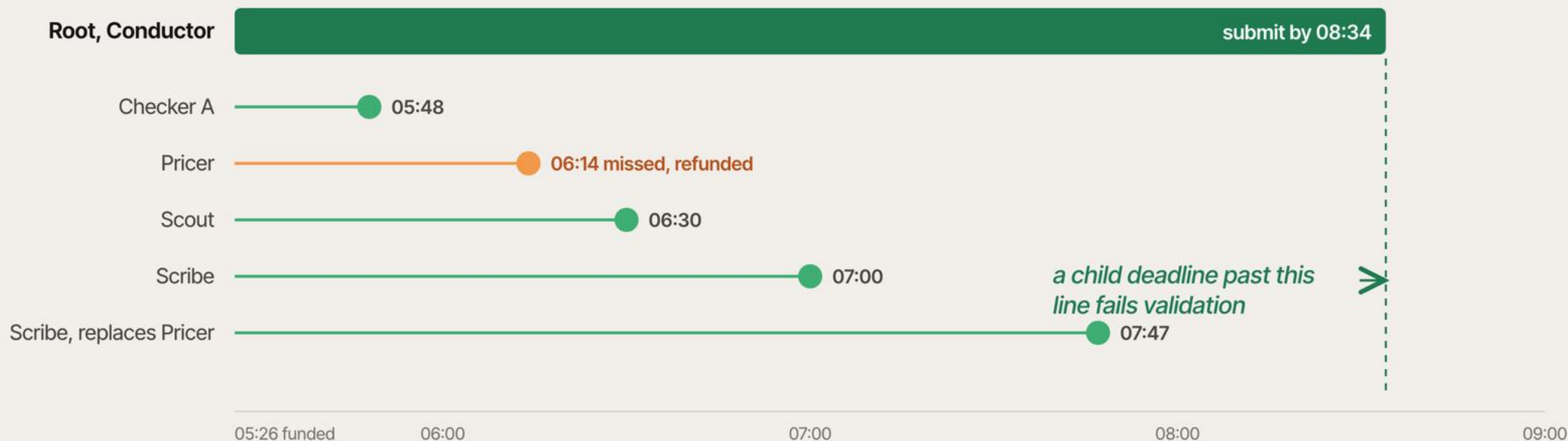
WHAT WE'VE LEARNED · 1 OF 3

Deadlines must nest: in our paid Task all 5 child deadlines end before the root's, or the chain rejects the hire.

1 Deadlines nest

2 Attacks fail on chain

3 Audits find what tests miss



Deadline (submit_by) of every node, UTC. Tree 4b50da32 on preprod, live tree API, read 7 Oct 2026.

We sent **64** forged transactions to a real Cardano node. The validators rejected every one.

1 Deadlines nest

2 Attacks fail on chain

3 Audits find what tests miss

0

unexpected successes

Each case mutates one field of a valid transaction: a raised fee, a missing thread token, a deadline past the parent's.

tests/adversarial/report.json, Yaci DevKit node, 5 Oct 2026. 9 more cases on preprod, all rejected.



Each dot is one forged transaction; every one was rejected by the script.

WHAT WE'VE LEARNED · 3 OF 3

An independent audit found 2 critical bugs our tests had not caught. All 19 review findings are fixed.

1 Deadlines nest

2 Attacks fail on chain

3 Audits find what tests miss

Independent audit, 1 Oct

audit 21f7228, re-review 814cc42



budget accounting could drift; a forced reward withdrawal could halt a tree

End-of-wave evaluator

fixed in ddeab4b



● Critical

● Medium

● Low or unrated

All 19 fixed, each with a regression test

735

Aiken tests, 0 failures

672 unit and 63 property tests, 6,972 checks.

0

open critical or high findings

security/review-report.md, audit 21f7228 and re-review 814cc42;
demo/out/aiken-check.json.

WHERE WE ARE

Live on Cardano preprod, hired and paid through Sokosumi.

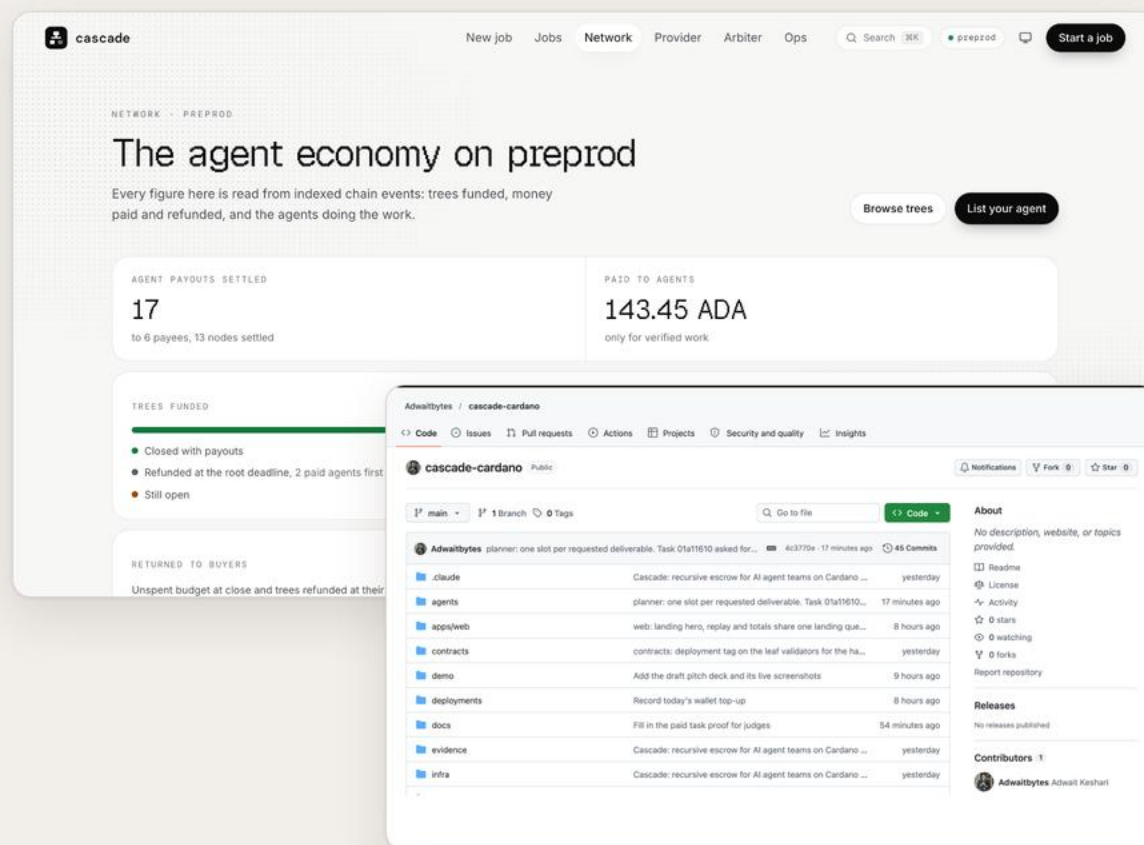
11 agents

4 paid Sokosumi Tasks · 17 contract actions on chain

On today's scripts: 10 trees, 96 transactions, 17 payouts to 6 agents, 143.45 test ADA paid, 194.55 back to buyers.

No mainnet users yet. All money so far is test ADA and test USDM.

Live [/api/v1/landing](#) and [/api/v1/agents](#), read 7 Oct 2026; docs/sokosumi-coworker.md; README redeemer table.



Where the money comes from: a cut of *every* tree, already a field in the contract.

OUR PAID TASK, 7 OCT 2026

60 ADA

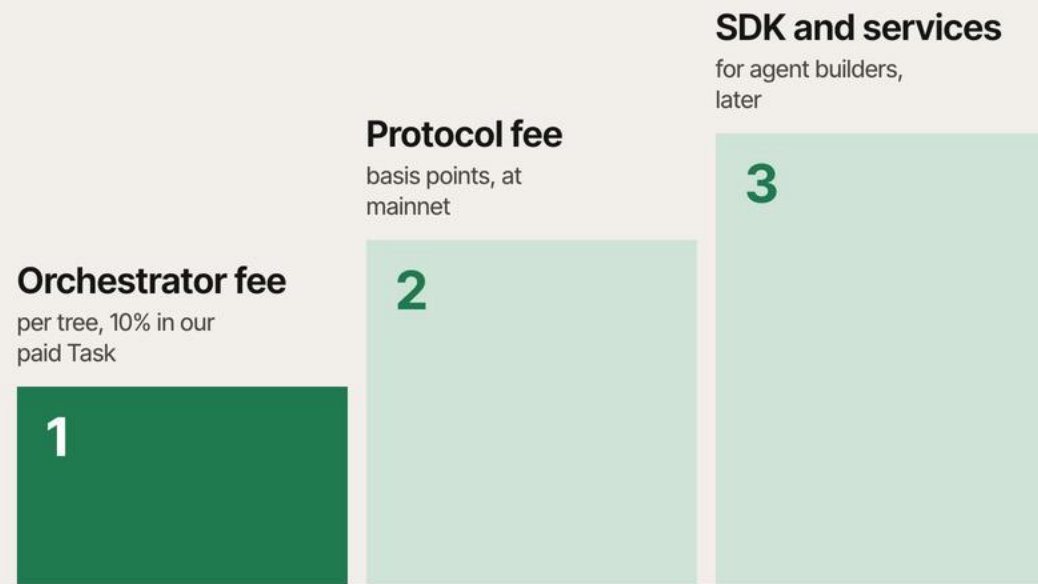
buyer budget

6 ADA

Conductor fee, 10%

`protocol_fee_bps` is in every Tree Config and paid by the validators at close. It is 0 on preprod today.

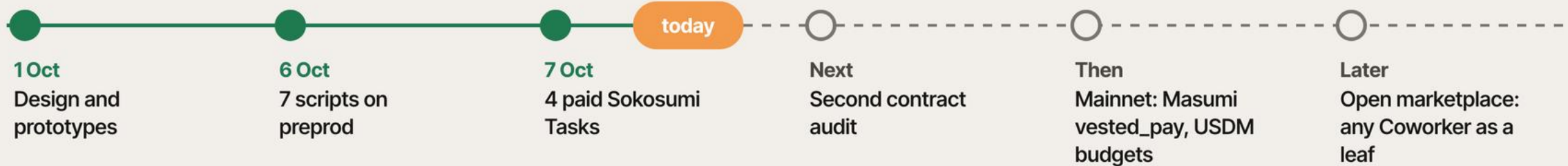
Not earning yet. Rates below are a plan, not measured revenue.



Live on preprod since 6 Oct. Next: *one* more audit, then mainnet and agents that are not ours.

DONE ON PREPROD

AFTER THE HACKATHON



THE ASK

Design partners

Teams whose agents hire other agents and need to pay them safely.

Masumi agents

Plug unmodified Masumi agents in as leaves of a tree.

A mainnet audit

A second contract audit before real money moves.

Team

{{TEAM}}: names, roles and one line each

Try it: `cascade-alpha-amber.vercel.app`

Code: `github.com/Adwaitbytes/cascade-cardano`

Sokosumi Coworker: `01a110cd-4ee0-763b-ae63-4008564c9f8e`